

IMAT MOCK EXAM 11 (Legacy): Final Dress Rehearsal

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 42/90.0 · Mock exam

Section A - Reading Skills, General Knowledge & Logical Reasoning (Q1-9)

1. Memories formed during the day are initially fragile. In deep sleep, patterns of activity recorded in the hippocampus replay the firing sequences that occurred while a task was being learned, and this replay coincides with brief rhythmic events that coordinate communication between the hippocampus and the cortex. In rodents, experimentally disrupting these replay events selectively impairs later performance on the learned task while leaving performance on other tasks intact. In humans, volunteers kept awake after memorising a list of words recall fewer items the next day than volunteers allowed to sleep, even when both groups are tested at the same interval after learning. On this evidence, sleep is not a passive interruption of mental life but an active part of how new memories become stable.

Which of the following statements is most strongly supported by the passage?

- A)** Staying awake after memorising a word list has no effect on next-day recall in humans.
- B)** People kept awake after learning remember more because less time has passed since learning.
- C)** Disrupting hippocampal replay impairs rodents' performance on every task that is tested.
- D)** All human memories are formed during deep sleep.
- E)** Sleep appears to play an active role in stabilising newly formed memories.

2. Wind and solar generators share a property that conventional power plants do not: their output follows the weather rather than demand. On windy spring afternoons a grid may receive more electricity than it can use, while on still winter evenings, when demand peaks, renewable output can fall close to zero. Grid-scale batteries installed in recent years address part of this mismatch by shifting surplus electricity across a few hours, and they have become cheap enough to be deployed in large numbers. They cannot, however, shift a summer surplus to a winter deficit, because storing electricity for months remains far too expensive with current technology. Electricity systems that rely heavily on wind and solar therefore still need some combination of long-duration storage, flexible demand, deliberately overbuilt generation and backup plants that can be counted on whatever the weather.

Which of the following statements is most strongly supported by the passage?

- A)** Flexible demand on its own can replace all backup generation in a renewable-based system.
- B)** The output of wind and solar plants follows electricity demand more closely than that of conventional plants.
- C)** Because batteries have become cheap, overbuilding renewable generation is no longer necessary.
- D)** Batteries can help balance renewable supply and demand over periods of a few hours, but not over whole seasons.
- E)** Grid-scale batteries are already able to store summer surplus electricity for use in winter.

3. (General knowledge) The Italian physician Camillo Golgi, who spent most of his career at the University of Pavia, shared the 1906 Nobel Prize in Physiology or Medicine, awarded for work on the structure of the nervous system. With which scientist did he share it?

- A)** Ivan Pavlov
- B)** Robert Koch
- C)** Santiago Ramón y Cajal
- D)** Emil von Behring
- E)** Charles Sherrington

4. (General knowledge) At the Milano Cortina 2026 Winter Olympic Games, held in Italy in February 2026, which country finished at the top of the final medal table?

- A)** Norway **B)** United States **C)** Netherlands **D)** Germany **E)** Italy

- 5.** Some buses in the city fleet are electric. All electric vehicles are exempt from the congestion charge. No vehicle that is exempt from the congestion charge is allowed to use the toll road. Which one of the following statements must be true?
- A)** Some electric buses in the city fleet use the toll road.
 - B)** Some buses in the city fleet are not allowed to use the toll road.
 - C)** No bus in the city fleet is allowed to use the toll road.
 - D)** The buses in the city fleet that are not electric must pay the congestion charge.
 - E)** Every vehicle that is exempt from the congestion charge is electric.
- 6.** Five runners, Aisha, Bruno, Carmen, Devi and Lena, finished a 400 metre race, taking places 1 to 5 with no ties. Lena finished third. Carmen finished immediately behind Lena. Devi finished behind Carmen. Bruno did not finish first. Which one of the following statements must be true?
- A)** Bruno finished ahead of Aisha.
 - B)** Bruno finished second.
 - C)** Aisha finished third.
 - D)** Devi finished fourth.
 - E)** Carmen finished immediately behind Devi.
- 7.** At a summer camp, each of the 100 children does at least one of three activities: swimming, sailing and archery. In total, 55 swim, 50 sail and 45 do archery; 25 swim and sail, 20 swim and do archery, 15 sail and do archery; and 10 do all three. How many children do exactly two of the three activities?
- A)** 20 **B)** 25 **C)** 30 **D)** 35 **E)** 40
- 8.** 1 April 2026 was a Wednesday. What was the date of the second Monday of April 2026?
- A)** 5 April **B)** 6 April **C)** 12 April **D)** 13 April **E)** 20 April
- 9.** A cube is folded from a flat net of six identical numbered squares. The net consists of one column of four squares numbered 5, 1, 3 and 6 from top to bottom, with the square numbered 2 attached to the left edge of square 1 and the square numbered 4 attached to the right edge of square 1. When the net is folded into a cube, which number is on the face opposite the face numbered 1?
- A)** 2 **B)** 3 **C)** 4 **D)** 5 **E)** 6

Section B - Biology (Q10-32)

- 10.** Which statement about the enzymes and DNA synthesis at a bacterial replication fork is correct?
- A)** The leading strand is synthesised continuously from a single primer, while the lagging strand is synthesised in short fragments as the fork opens.
 - B)** The leading strand requires a new primer for each fragment it synthesises.
 - C)** DNA ligase unwinds the parental double helix ahead of the replication fork.
 - D)** Helicase synthesises the short RNA primers needed to start DNA synthesis.
 - E)** DNA polymerase can begin synthesis of a new strand without needing a primer.
- 11.** Which statement about transcription in a eukaryotic cell is correct?
- A)** Transcription begins when RNA polymerase binds to the terminator sequence of a gene.
 - B)** RNA polymerase synthesises the new RNA strand in the 3' to 5' direction.
 - C)** The poly-A tail is a stretch of DNA sequence copied from the end of the gene.
 - D)** RNA polymerase binds to the promoter and starts RNA synthesis without a primer, using one DNA strand as a template.
 - E)** RNA polymerase requires an RNA primer before it can start copying a gene.

- 12.** Which statement about the ribosome during translation is correct?
- A)** In eukaryotes, ribosomes begin translating an mRNA while that mRNA is still being transcribed.
 - B)** A tRNA carrying the next amino acid enters at the A site, where its anticodon pairs with the mRNA codon.
 - C)** The initiator tRNA binds first to the A site of the ribosome when translation starts.
 - D)** The E site is where peptide bonds are formed between amino acids.
 - E)** The growing polypeptide chain is carried on a tRNA sitting in the A site while new tRNAs enter at the P site.
- 13.** The beginning of an mRNA reads 5' AUG GCA UAC GGU UAA 3', where AUG is the start codon and UAA is a stop codon. Codon meanings: AUG = methionine, GCA = alanine, GCC = alanine, GUA = valine, UAC = tyrosine, GGU = glycine, CGG = arginine, UUA = leucine. Two independent mutations occur in different copies of this gene. Mutation 1 substitutes one base so that codon 2 reads GCC instead of GCA. Mutation 2 inserts a single G immediately after codon 2. Which statement about the resulting proteins is correct?
- A)** Mutation 2 adds exactly one extra amino acid to the protein and leaves all the other amino acids unchanged.
 - B)** Both mutations are silent, because neither changes any amino acid in the protein.
 - C)** Mutation 1 is silent, while mutation 2 shifts the reading frame, so from codon 3 onward the mRNA reads GUA, CGG and UUA instead of UAC, GGU and UAA, changing the amino acids and losing the stop codon.
 - D)** Mutation 1 changes the amino acid at position 2 of the protein, while mutation 2 is silent.
 - E)** Mutation 1 is silent and mutation 2 changes only the amino acid encoded by codon 2.
- 14.** In a testcross, a pea plant heterozygous for three independently assorting genes, AaBbCc, is crossed with a plant that is aabbcc. What is the probability that an offspring is heterozygous at all three loci (AaBbCc)?
- A)** 1/16 **B)** 1/8 **C)** 1/4 **D)** 1/2 **E)** 3/4
- 15.** Cystic fibrosis is caused by an autosomal recessive allele. Two parents without the condition have one child with cystic fibrosis and a second child who is healthy. What is the probability that this healthy second child is a carrier of the allele?
- A)** 1/4 **B)** 1/3 **C)** 1/2 **D)** 2/3 **E)** 3/4
- 16.** Haemophilia is caused by a recessive allele on the X chromosome. A woman has no symptoms, but her father had haemophilia. She and her husband, who does not have haemophilia, are expecting their first child. What is the probability that the first child is a daughter who carries the allele?
- A)** 1/8 **B)** 1/4 **C)** 1/2 **D)** 1 **E)** 3/4
- 17.** In a series of crosses, the recombination frequencies between three linked genes are: between gene a and gene b, 4 percent; between gene b and gene c, 9 percent; between gene a and gene c, 13 percent. Which conclusion is correct?
- A)** The genes are arranged with gene b between the other two, in the order a b c.
 - B)** The genes are arranged with gene a between the other two, in the order b a c.
 - C)** No valid order exists, because the recombination frequency between two genes must be the same whichever third gene is considered.
 - D)** The genes are arranged with gene c between the other two, in the order a c b.
 - E)** The three genes must lie on three different chromosomes.
- 18.** Which event occurs during anaphase of mitosis in an animal cell?
- A)** Nuclear envelopes re-form around each group of chromosomes.
 - B)** Chromosomes align along the equator of the cell.
 - C)** The nuclear envelope breaks down and the spindle begins to form.
 - D)** Homologous chromosome pairs separate and move to opposite poles.
 - E)** Sister chromatids separate at the centromeres and move toward opposite poles.

- 19.** During meiosis II in a human oocyte, the sister chromatids of chromosome 21 fail to separate, while meiosis I and the rest of meiosis II proceed normally. One of the resulting egg cells receives both copies of chromosome 21 and is then fertilised by a normal sperm. Which statement is correct?
- A)** The zygote has 45 chromosomes, with only one copy of chromosome 21.
 - B)** The zygote has 69 chromosomes because all chromosomes are present in triplicate.
 - C)** The zygote has 47 chromosomes, with three copies of chromosome 21, the condition known as Down syndrome.
 - D)** The zygote has 46 chromosomes and is chromosomally normal.
 - E)** The zygote has 47 chromosomes, but the extra chromosome is a sex chromosome.
- 20.** Before industrialisation, light coloured peppered moths rested on pale, lichen covered tree trunks and greatly outnumbered dark coloured moths. As industrial soot blackened the trunks, the dark form became far more common within a few decades. Which statement best explains this change?
- A)** The dark form arose by random mutation before the trunks darkened; once the trunks became dark, birds caught more light moths, so dark moths survived and reproduced more successfully and passed on their alleles.
 - B)** Soot chemically altered the genes of the moths so that their offspring were born dark.
 - C)** Individual moths gradually darkened during their own lifetimes as soot settled on the trees.
 - D)** The moths sensed the darkening trunks and produced dark offspring in order to blend into the background.
 - E)** Dark coloured moths were already more common than light moths before the trees darkened.
- 21.** A population of diploid flowering plants grows in one meadow with no physical barriers. A mutation arises in one lineage that doubles the chromosome number, producing tetraploid plants. Tetraploids mate successfully with other tetraploids, but crosses between tetraploid and diploid plants produce seeds that die before germination. Which statement is correct?
- A)** A change in chromosome number can alter gene expression but can never give rise to a new species.
 - B)** This is allopatric speciation, because the tetraploid and diploid plants cannot interbreed.
 - C)** The tetraploid and diploid plants still belong to one biological species, because they live in the same meadow.
 - D)** No speciation can occur without a physical barrier such as a river or a mountain range.
 - E)** This is sympatric speciation: a reproductively isolated polyploid lineage has formed within the same geographic area.
- 22.** Which statement about the role of water in the light dependent reactions of photosynthesis is correct?
- A)** Electrons released by the splitting of water are passed directly to photosystem I to replace its lost electrons.
 - B)** The oxygen released by photosynthesis comes from carbon dioxide that is split in the thylakoid membranes.
 - C)** The oxygen released by photosynthesis comes from water molecules split during the light dependent reactions.
 - D)** Water is split in the stroma as part of the Calvin cycle.
 - E)** The splitting of water directly produces the ATP that the Calvin cycle consumes.
- 23.** Which statement about long distance transport in flowering plants is correct?
- A)** Xylem transports water and dissolved mineral ions upward from the roots, while phloem transports sugars from source tissues to sink tissues.
 - B)** The contents of both xylem and phloem are moved mainly by the transpiration pull from the leaves.
 - C)** Phloem can transport sugars only downward, from the leaves to the roots.
 - D)** Mature xylem vessels consist of living cells that actively pump water from cell to cell.
 - E)** Xylem transports sugars from the leaves to the roots, while phloem transports water and mineral ions upward.

- 24.** In a healthy heart, the ventricles begin to contract shortly after the atria contract. Which statement explains this timing?
- A)** The atria and the ventricles are excited at the same moment, but ventricular muscle contracts more slowly than atrial muscle.
 - B)** The ventricles contract first and the atria contract during the diastole that follows.
 - C)** A second impulse from the sinoatrial node travels directly to the ventricles and triggers their contraction.
 - D)** The delay is caused by the time blood takes to flow from the atria into the ventricles.
 - E)** The impulse from the sinoatrial node is delayed at the atrioventricular node, so the ventricles are excited shortly after the atria, allowing the atria to finish emptying before the ventricles contract.
- 25.** Which statement about ultrafiltration at the glomerulus is correct?
- A)** Plasma proteins pass freely into the filtrate and are reabsorbed later along the loop of Henle.
 - B)** Red blood cells normally enter the filtrate and are returned to the blood in the distal tubule.
 - C)** Glucose is actively transported from the blood into the filtrate at the glomerulus.
 - D)** Fluid is forced from the glomerular capillaries into Bowman's capsule by blood pressure, and the filtrate normally contains glucose, urea and ions but no plasma proteins or blood cells.
 - E)** Water is moved out of the glomerular capillaries by active transport.
- 26.** Which statement about nerve action potentials is correct?
- A)** Subthreshold stimuli produce small action potentials that travel more slowly along the axon.
 - B)** Once threshold is reached, the amplitude of the action potential is fixed; a stronger stimulus increases the frequency of action potentials, not their size.
 - C)** A stronger stimulus produces a single action potential with a larger amplitude.
 - D)** During repolarisation, the inside of the axon becomes even more positive than at the peak of depolarisation.
 - E)** The refractory period allows action potentials to travel backwards along the axon as well as forwards.
- 27.** Which statement about the control of cortisol secretion is correct?
- A)** Cortisol raises its own secretion through a positive feedback loop.
 - B)** When cortisol levels fall, the anterior pituitary releases less ACTH so that cortisol stays low.
 - C)** ACTH travels down nerve fibres from the hypothalamus to the adrenal cortex.
 - D)** A rise in blood cortisol acts by negative feedback on the hypothalamus and anterior pituitary, reducing the release of CRH and ACTH.
 - E)** Cortisol is released by the adrenal medulla in response to nerve signals from the hypothalamus.
- 28.** Which statement about B and T lymphocytes is correct?
- A)** Both B and T lymphocytes complete their maturation in the thymus.
 - B)** T lymphocytes are the main antibody secreting cells of the immune system.
 - C)** Activated B lymphocytes differentiate into plasma cells that secrete antibodies, whereas cytotoxic T cells destroy infected body cells by direct contact.
 - D)** Cytotoxic T cells engulf bacteria by phagocytosis and digest them internally.
 - E)** B lymphocytes mature in the thymus before circulating in the blood.
- 29.** Which statement about the cytoskeleton is correct?
- A)** Microfilaments are hollow tubes of tubulin that give cilia their characteristic internal arrangement.
 - B)** Microtubules are permanently fixed structures that never assemble or disassemble inside the cell.
 - C)** Actin microfilaments form the spindle fibres that separate chromosomes during mitosis.
 - D)** The contractile ring that divides an animal cell during cytokinesis is made of microtubules.
 - E)** Microtubules are hollow polymers of tubulin that form the mitotic spindle and the core of cilia and flagella.

- 30.** Which statement about apoptosis is correct?
- A)** Apoptosis is a genetically regulated process in which the cell is dismantled into fragments that nearby cells engulf without causing inflammation.
 - B)** Cells dying by apoptosis release their digestive enzymes into the surrounding tissue.
 - C)** Apoptosis plays no role in healthy embryonic development.
 - D)** Apoptosis is triggered only by external toxins and never by signals arising within the cell itself.
 - E)** Apoptosis is the uncontrolled death of cells after injury, and it always triggers inflammation.
- 31.** Which statement correctly matches a level of protein structure with what defines it?
- A)** The primary structure is defined by hydrogen bonding between atoms of the polypeptide backbone.
 - B)** The tertiary structure is the assembly of several separate polypeptide chains into one functional unit.
 - C)** The secondary structure is defined by disulfide bridges between cysteine residues.
 - D)** The primary structure is the sequence of amino acids joined together by peptide bonds.
 - E)** The quaternary structure is the local coiling of the chain into alpha helices.
- 32.** A person standing outside in cold weather starts to shiver. Which statement about this response is correct?
- A)** Shivering is involuntary contraction of skeletal muscles that generates heat, coordinated by the hypothalamus as part of a negative feedback response to falling body temperature.
 - B)** Skin blood vessels widen so that warm blood flows close to the surface, warming the skin.
 - C)** Sweat production increases so that evaporation can release heat from the body core.
 - D)** Shivering is a voluntary response that the person chooses to start in order to warm up.
 - E)** Shivering cools the body further, because the contracting muscles push warm blood to the skin surface where heat is lost.

Section C - Chemistry (Q33-47)

- 33.** What is the mass of 0.25 mol of copper(II) sulfate, CuSO_4 ? (Cu = 64, S = 32, O = 16)
- A)** 32 g **B)** 40 g **C)** 64 g **D)** 160 g **E)** 640 g
- 34.** 28 g of nitrogen gas (N_2 , M = 28 g/mol) is mixed with 9.0 g of hydrogen gas (H_2 , M = 2.0 g/mol) and the mixture is reacted according to: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ (NH_3 , M = 17 g/mol). What is the maximum mass of ammonia that can be produced?
- A)** 2.0 g **B)** 17 g **C)** 34 g **D)** 51 g **E)** 68 g
- 35.** An oxide of sulfur is found by analysis to contain 50.0% sulfur and 50.0% oxygen by mass. What is its empirical formula? (S = 32, O = 16)
- A)** S_2O **B)** SO **C)** SO_4 **D)** SO_3 **E)** SO_2
- 36.** 5.85 g of sodium chloride (M = 58.5 g/mol) is dissolved in water and the solution is made up to exactly 500 cm^3 . What is the concentration of the solution?
- A)** 0.050 mol/ dm^3 **B)** 0.10 mol/ dm^3 **C)** 0.20 mol/ dm^3 **D)** 0.40 mol/ dm^3 **E)** 0.50 mol/ dm^3
- 37.** Potassium hydroxide is a strong base that dissociates completely in water. What is the pH at 25 °C of a 1.0×10^{-4} mol/ dm^3 solution of KOH? ($K_w = 1.0 \times 10^{-14}$)
- A)** 4 **B)** 7 **C)** 10 **D)** 11 **E)** 14
- 38.** Hydrochloric acid is titrated with sodium hydroxide solution. Which of the following statements is correct?
- A)** At the equivalence point neither reactant is in excess; since the salt NaCl does not hydrolyse, the solution has pH 7 at 25 °C.
 - B)** The equivalence point is defined as the point at which the indicator changes colour.
 - C)** At the equivalence point the acid is still in excess, so the pH is below 7.
 - D)** At the equivalence point, equal masses of acid and base have been added.
 - E)** At the equivalence point of any titration, weak or strong, the pH is exactly 7.

- 39.** In acidic solution, permanganate ions oxidise ethanedioate (oxalate) ions: $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$. When the equation is balanced with the smallest whole-number coefficients, what is the coefficient of H^+ ?
- A)** 2 **B)** 5 **C)** 10 **D)** 16 **E)** 32
- 40.** Which of the following statements about the octet rule is correct?
- A)** BF_3 violates the octet rule because boron carries more than eight valence electrons.
B) Boron in BF_3 has only six electrons in its valence shell, so BF_3 can accept an electron pair and act as a Lewis acid.
C) Phosphorus in PCl_5 obeys the octet rule, because each chlorine atom shares one electron with it.
D) Compounds of period 3 elements such as PCl_5 and SF_6 cannot exist because they exceed the octet.
E) Nitrogen in NH_3 has an expanded octet of ten electrons.
- 41.** Which of the following statements about the halogens (Group 17) is correct?
- A)** Bromine displaces chlorine from potassium chloride solution, because bromine is the stronger oxidising agent.
B) Their boiling points decrease down the group, so astatine is a gas.
C) Their reactivity increases down the group because the outer shell is further from the nucleus.
D) They are all good electrical conductors in the liquid state.
E) Their reactivity decreases down the group because the atoms get larger and attract an additional electron less strongly.
- 42.** The standard enthalpies of formation of methane $\text{CH}_4(\text{g})$, carbon dioxide $\text{CO}_2(\text{g})$ and liquid water $\text{H}_2\text{O}(\text{l})$ are -75, -394 and -286 kJ/mol respectively. What is the enthalpy change for the complete combustion of one mole of methane: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$? (The enthalpy of formation of $\text{O}_2(\text{g})$ is zero.)
- A)** -1041 kJ/mol **B)** -891 kJ/mol **C)** -605 kJ/mol **D)** +605 kJ/mol **E)** +891 kJ/mol
- 43.** Hydrogen and iodine reach equilibrium: $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$. At equilibrium the concentrations are $[\text{H}_2] = 0.20 \text{ mol/dm}^3$, $[\text{I}_2] = 0.30 \text{ mol/dm}^3$ and $[\text{HI}] = 0.60 \text{ mol/dm}^3$. What is the value of K_c ?
- A)** 0.17 **B)** 1.2 **C)** 2.45 **D)** 6.0 **E)** 10
- 44.** For the reaction $\text{A} + \text{B} \rightarrow \text{products}$ at 25°C , the following initial rates were measured. Experiment 1: $[\text{A}] = 0.10 \text{ mol/dm}^3$, $[\text{B}] = 0.10 \text{ mol/dm}^3$, rate = $0.0020 \text{ mol/(dm}^3 \text{ s)}$. Experiment 2: $[\text{A}] = 0.20 \text{ mol/dm}^3$, $[\text{B}] = 0.10 \text{ mol/dm}^3$, rate = $0.0040 \text{ mol/(dm}^3 \text{ s)}$. Experiment 3: $[\text{A}] = 0.10 \text{ mol/dm}^3$, $[\text{B}] = 0.20 \text{ mol/dm}^3$, rate = $0.0080 \text{ mol/(dm}^3 \text{ s)}$. What is the rate law for this reaction?
- A)** rate = $k[\text{A}][\text{B}]^2$
B) rate = $k[\text{A}]^2[\text{B}]^2$
C) rate = $k[\text{B}]^2$
D) rate = $k[\text{A}][\text{B}]$
E) rate = $k[\text{A}]^2[\text{B}]$
- 45.** An alkane consists of a chain of five carbon atoms, with one methyl group attached to the second carbon of the chain and another methyl group attached to the fourth carbon. What is the IUPAC name of this alkane?
- A)** 2-methylhexane
B) 2,3-dimethylpentane
C) 2,4-dimethylpentane
D) 2,4-dimethylhexane
E) 3,5-dimethylpentane

46. Methane reacts with chlorine gas in sunlight: $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$, followed by further substitution steps. What type of reaction is the first of these steps, $\text{CH}_4 \rightarrow \text{CH}_3\text{Cl}$?

- A)** an electrophilic addition
- B)** an elimination
- C)** a nucleophilic substitution
- D)** a condensation
- E)** a free-radical substitution

47. What is the electron configuration of an iron(II) ion, Fe^{2+} ? (Iron has atomic number 26.)

- A)** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$
- B)** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$
- C)** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
- D)** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$
- E)** $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$

Section D - Mathematics (Q48-54)

48. Solve the equation $\log_2(x) + \log_2(x - 6) = 4$.

- A)** $x = 16$
- B)** There is no solution
- C)** $x = 8$ or $x = -2$
- D)** $x = -2$ only
- E)** $x = 8$ only

49. A committee of 3 people is chosen at random from a group of 4 chemists and 6 physicists. What is the probability that the committee contains exactly one chemist?

- A)** $1/6$ **B)** $1/2$ **C)** $3/10$ **D)** $3/5$ **E)** $5/6$

50. If $f(x) = x^2 - 4x$, which of the following expressions is equal to $f(x + 3) - 2$ for every value of x ?

- A)** $x^2 - 4x + 1$ **B)** $x^2 - 4x - 5$ **C)** $x^2 + 2x - 1$ **D)** $x^2 + 2x - 5$ **E)** $x^2 - 10x + 19$

51. How many values of x satisfy the equation $\cos(2x) = 1/2$ in the interval $0 \leq x \leq 2\pi$?

- A)** 1 **B)** 2 **C)** 3 **D)** 4 **E)** 6

52. Solve the inequality $(x + 1)/(x - 2) \geq 0$.

- A)** $-1 \leq x < 2$
- B)** $x \leq -1$ or $x \geq 2$
- C)** $x \geq -1$ and $x < 2$
- D)** $x < -1$ or $x > 2$
- E)** $x \leq -1$ or $x > 2$

53. A square is inscribed in a circle of radius 6 cm. What is the area of the square?

- A)** 24 cm^2 **B)** 36 cm^2 **C)** 72 cm^2 **D)** 108 cm^2 **E)** 144 cm^2

54. In an arithmetic sequence the third term is 11 and the eighth term is 31. What is the sum of the first ten terms?

- A)** 105 **B)** 210 **C)** 290 **D)** 390 **E)** 420

Section E - Physics (Q55-60)

55. A ball is kicked from level ground with an initial speed of 20 m/s at an angle of 30 degrees above the horizontal. What is the maximum height it reaches? ($g = 10 \text{ m/s}^2$; $\sin 30 \text{ degrees} = 0.5$; $\cos 30 \text{ degrees} = 0.87$; ignore air resistance.)

- A)** 2.5 m **B)** 5 m **C)** 10 m **D)** 20 m **E)** 40 m

56. A ball of mass 4.0 kg rolls along level ground with kinetic energy 200 J. What is its speed?

- A)** 5 m/s **B)** 10 m/s **C)** 25 m/s **D)** 50 m/s **E)** 100 m/s

- 57.** On a frictionless horizontal track a trolley of mass 2.0 kg moving at 6.0 m/s collides head-on and elastically with a stationary trolley of mass 4.0 kg. What is the speed of the 4.0 kg trolley immediately after the collision?
- A)** 2 m/s **B)** 3 m/s **C)** 4 m/s **D)** 6 m/s **E)** 12 m/s
- 58.** An object of weight 50 N and density 2500 kg/m³ hangs fully submerged in water of density 1000 kg/m³, held at rest by a vertical string. What is the tension in the string? ($g = 10 \text{ m/s}^2$.)
- A)** 12.5 N **B)** 20 N **C)** 30 N **D)** 50 N **E)** 70 N
- 59.** How much heat is needed to raise the temperature of 0.30 kg of olive oil from 20 degrees C to 70 degrees C? The specific heat capacity of the oil is 2000 J/(kg K); ignore heat losses.
- A)** 3.0 kJ **B)** 15 kJ **C)** 30 kJ **D)** 60 kJ **E)** 120 kJ
- 60.** A 12 V battery of negligible internal resistance is connected directly across a single 8 Ω resistor. What is the power dissipated in the resistor?
- A)** 1.5 W **B)** 18 W **C)** 24 W **D)** 96 W **E)** 144 W

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	E	2	D	3	C	4	A	5	B
6	B	7	C	8	D	9	E	10	A
11	D	12	B	13	C	14	B	15	D
16	B	17	A	18	E	19	C	20	A
21	E	22	C	23	A	24	E	25	D
26	B	27	D	28	C	29	E	30	A
31	D	32	A	33	B	34	C	35	E
36	C	37	C	38	A	39	D	40	B
41	E	42	B	43	D	44	A	45	C
46	E	47	A	48	E	49	B	50	D
51	D	52	E	53	C	54	B	55	B
56	B	57	C	58	C	59	C	60	B